Grouped frequency tables and histograms

1

Class	Frequency
$30 - 39$	3
$40 - 49$	5
$50 - 59$	3
$60 - 69$	8
$70 - 79$	2
$80 - 89$	4

2

Class	Class centre (x)	Frequency (f)	xf
$21 - 25$	23	3	69
$26 - 30$	28	4	112
$31 - 35$	33	2	66
$36 - 40$	38	8	304
$41 - 45$	43	3	129
$46 - 50$	48	5	240

3

Class	Class centre (x)	Frequency (f)	xf
$11 - 15$	13	6	78
$16 - 20$	18	8	144
$21 - 25$	23	2	46
$26 - 30$	28	2	56
$31 - 35$	33	5	165
$36 - 40$	38	1	38

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4

a

Class	Class centre (x)	f	xf
$1-9$	5	8	40
$10-18$	14	6	84
$19-27$	23	4	92
$28-36$	32	6	192
$37-45$	41	8	328
Total		32	736

b 23.00

5 $30-39$

6 23.0

7

a 9.3

b $1-5$

8

a Ungrouped

b Grouped

9

a 33.5

b 21

10 $19 < \text{height} \leq 28$

11

a 7

b 4

12

a

Score (x)	Frequency (f)
$10-19$	3
$20-29$	4
$30-39$	2
$40-49$	1
$50-59$	1

b

Score (x)	Frequency (f)
$30-39$	5
$40-49$	4
$50-59$	2
$60-69$	8
$70-79$	3
$80-89$	3

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13

a $20 \to 29, 30 \to 39, 40 \to 49, 50 \to 59,$ and $60 \to 69$

b

Score (x)	$20 \to 29$	$30 \to 39$	$40 \to 49$	$50 \to 59$	$60 \to 69$
Frequency (f)	3	2	1	3	2

14

a $10 \to 19, 20 \to 29, 30 \to 39, 40 \to 49, 50 \to 59, 60 \to 69,$ and $70 \to 79$

b

Score (x)	Frequency (f)
$10 \to 19$	7
$20 \to 29$	3
$30 \to 39$	2
$40 \to 49$	3
$50 \to 59$	6
$60 \to 69$	4
$70 \to 79$	4

15

a

Score (x)	Frequency (f)
$1 - 5$	6
$6 - 10$	12
$11 - 15$	7
$16 - 20$	3
$21 - 25$	9
$26 - 30$	2
$31 - 35$	4

b

Score (x)	Frequency (f)
$20 - 29$	150
$30 - 39$	140
$40 - 49$	160
$50 - 59$	150
$60 - 69$	130
$70 - 79$	110
$80 - 89$	120
$90 - 99$	130

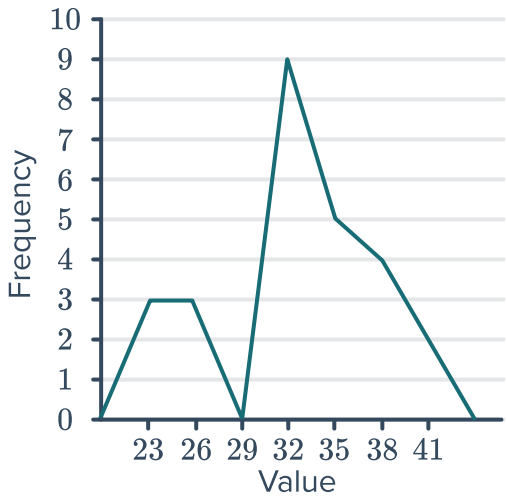
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16

a

Class interval	Class centre	Frequency
$21.5 \leq p < 24.5$	23	3
$24.5 \leq p < 27.5$	26	3
$27.5 \leq p < 30.5$	29	0
$30.5 \leq p < 33.5$	32	9
$33.5 \leq p < 36.5$	35	5
$36.5 \leq p < 39.5$	38	4
$39.5 \leq p < 42.5$	41	2

b



c 0

d 17

e 15

f 20

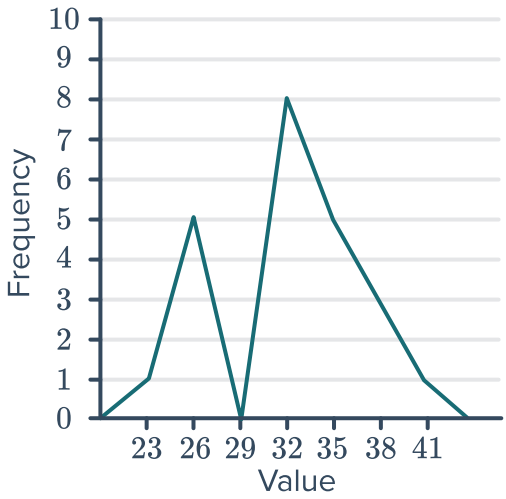
17

a

Class limits	Midpoint	Frequency
$21.5 - 24.5$	23	1
$24.5 - 27.5$	26	5
$27.5 - 30.5$	29	0
$30.5 - 33.5$	32	8
$33.5 - 36.5$	35	5
$36.5 - 39.5$	38	3
$39.5 - 42.5$	41	1

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b



c 0

d 18

e 14

f 17

Applications

18

a Yes **b** Yes **c** Yes **d** No

19

a

Heart Rate	Class Centre	Frequency
$45 - 54$	49.5	15
$55 - 64$	59.5	25
$65 - 74$	69.5	26
$75 - 84$	79.5	30

b 96 people

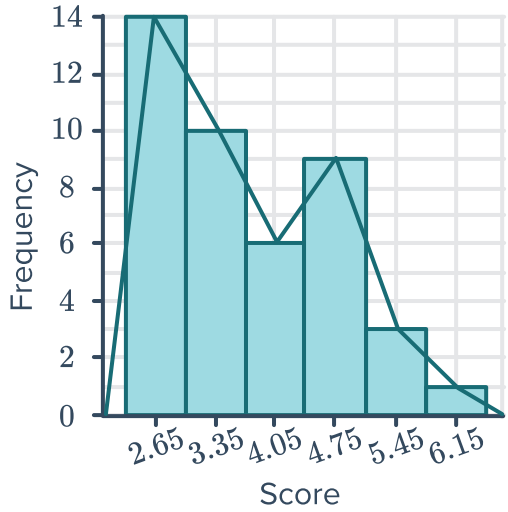
c 51 people

d 40 people

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20

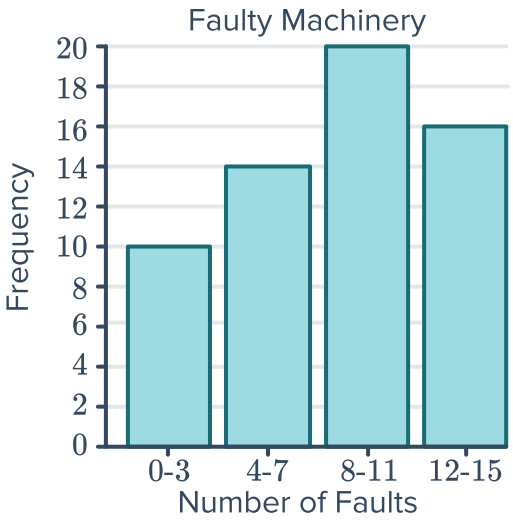
a



b The areas are the same.

21

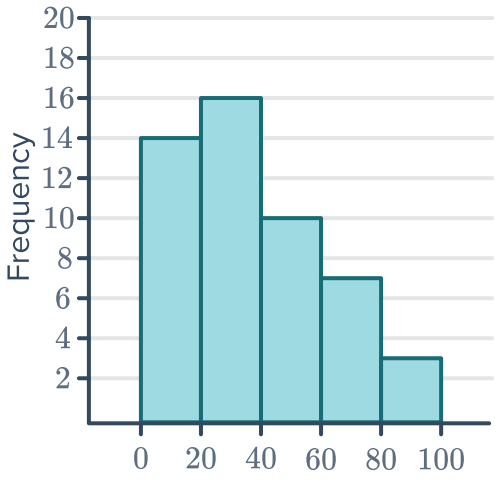
a



b 0

22

a



b

i No **ii** Yes **iii** No **iv** No

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23

a

i $0 \leq \text{rate} < 5$

ii $5 \leq \text{rate} < 10$

iii $10 \leq \text{rate} < 15$

iv $15 \leq \text{rate} < 20$

b 3

24

a 140.9 cents

b 11 days

25

a 25 students

b 13 students

c 16%

26

a

b 31.7%

Height (cm)	Frequency
$95 \leq h < 100$	1
$100 \leq h < 105$	3
$105 \leq h < 110$	2
$110 \leq h < 115$	12
$115 \leq h < 120$	19
$120 \leq h < 125$	15
$125 \leq h < 130$	4
$130 \leq h < 135$	4
Total:	60

27

a

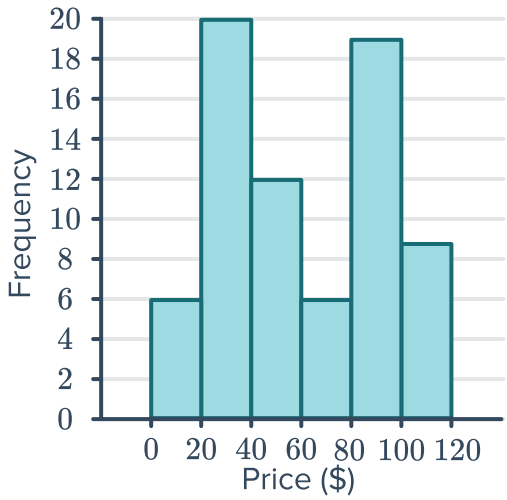
b 51.6%

Speed (km/h)	Frequency
$110 \leq s < 120$	4
$120 \leq s < 130$	12
$130 \leq s < 140$	8
$140 \leq s < 150$	5
$150 \leq s < 160$	2
Total:	31

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28

a



b $20 - 39$ and $80 - 99$

c The data shows that there are two main price ranges of books that were recently bought by university students: $\$20$ to $\$39$, and $\$80$ to $\$99$.

29

a

Battery life (minutes)	Frequency
$400 \leq x < 500$	3
$500 \leq x < 600$	9
$600 \leq x < 700$	13
$700 \leq x < 800$	32
$800 \leq x < 900$	22
$900 \leq x < 1000$	27
$1000 \leq x < 1100$	6
$1100 \leq x < 1200$	4
$1200 \leq x < 1300$	1
Total :	117

b 89.7%

c 9.4%

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30

a 35

b $60\ \text{mm} \leq x < 80\ \text{mm}$

c Yes

d No. The table shows ranges of lengths and not individual leaves. We do not know if any of the leaves in the range $0 \leq x < 20$ are less than $5\ \text{mm}$.

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